

Package ‘estbanr’

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Title Brazilian Monthly Banking Statistics by Municipality (ESTBAN)

Version 0.1.1

Description Download, read and tidy the ESTBAN (Estatística Bancária Mensal por Município, Monthly Banking Statistics by Municipality) files published by the Brazilian Central Bank (Banco Central do Brasil) for every bank branch and municipality in Brazil. Each file reports balance-sheet accounts of the COSIF (Plano Contábil das Instituições do Sistema Financeiro Nacional, the chart of accounts of the Brazilian financial system) such as credit operations, deposits and savings. Files are fetched from the official site <https://www.bcb.gov.br/estatisticas/estatisticabancariamunicipios> with an idempotent local cache, read from their Latin-1 encoded CSV (comma-separated values) layout into tibbles, optionally filtered by state, and aggregated by municipality. Includes tools to detect and impute institution-month non-reports (an institution present in the file with every account equal to zero), which would otherwise be mistaken for zero balances.

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URL <https://strategicprojects.github.io/estbanr/>,
<https://github.com/StrategicProjects/estbanr>

BugReports <https://github.com/StrategicProjects/estbanr/issues>

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Contents

estban_by_municipality	2
estban_cache_dir	3
estban_clean_names	4
estban_columns	5
estban_download	6
estban_fetch	7
estban_flag_nonreport	8
estban_impute_nonreport	9
estban_read	10
estban_url	11
Index	12

estban_by_municipality
<i>Aggregate ESTBAN to the municipality</i>

Description

Sums every account over the institutions (and branches) of each municipality and month. With `impute = TRUE` (the default) non-reported institution-months are interpolated first with `estban_impute_nonreport()`, so the municipal series does not collapse when a bank is missing from a file.

Usage

`estban_by_municipality(df, impute = TRUE, long = FALSE)`

Arguments

df	A tibble from <code>estban_read()</code> or <code>estban_fetch()</code> covering several months (interpolation needs neighbours). The <code>nonreport</code> column is computed when absent.
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impute	Logical. Interpolate non-reports before summing (default TRUE). With FALSE the raw rows are summed as published.
long	Logical. Return one row per municipality, month and account (verbete, value) instead of one column per account (default FALSE).

Value

A tibble keyed by uf, codmun_ibge, municipio and ref. In wide form the account columns follow; in long form the columns verbete (clean column name) and value follow.

Examples

```
f <- system.file("extdata", "202401_ESTBAN_AG_sample.CSV", package = "estbanr")
x <- estban_read(f)
m <- estban_by_municipality(x, impute = FALSE)
m[, c("uf", "municipio", "ref", "verbete_160_operacoes_de_credito",
      "verbete_420_depositos_de_poupanca")]

head(estban_by_municipality(x, impute = FALSE, long = TRUE))
```

estban_cache_dir	<i>Resolve the estbanr cache directory</i>
------------------	--

Description

The directory where downloaded ESTBAN files are kept. Resolution order:

Usage

```
estban_cache_dir(cache_dir = NULL, create = TRUE)
```

Arguments

cache_dir	Optional directory path. When NULL, the resolution order above applies.
create	Logical. Create the directory when it does not exist (default TRUE).

Details

1. the cache_dir argument;
2. the ESTBANR_CACHE_DIR environment variable;
3. the estbanr.cache_dir R option;
4. a session-scoped folder under `base::tempdir()` (the default, so the package never writes outside the temporary directory unless you opt in).

Set one of the persistent options (2 or 3) to keep files across sessions and avoid re-downloading months you already have.

estban_columns	<i>ESTBAN column layout and account dictionary</i>
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Description

`estban_columns()` lists the columns of the monthly file for a level, in file order, with their `snake_case` names. `estban_verbetes()` restricts the list to the balance-sheet accounts (`VERBETE_*`), parsing the three-digit 'COSIF' codes out of the column name. Some columns combine several accounts (for instance `VERBETE_141 + VERBETE_142`); those have more than one code in `codes` and `n_codes > 1`.

Usage

```
estban_columns(level = c("agencia", "municipio"))
```

```
estban_verbetes(level = c("agencia", "municipio"))
```

Arguments

`level` "agencia" (default) or "municipio".

Details

The dictionary is built from the January 2024 files (see `data-raw/verbetes.R` in the source repository) and covers both levels, which share the same accounts.

Value

A tibble. `estban_columns()` has `position`, `column` (original name), `name` (clean name) and `kind` ("id" or "verbeta"). `estban_verbetes()` has `column`, `name`, `codes` (character, codes separated by ;), `code` (first code, integer), `n_codes`, `side` ("ativo" for codes below 400, "passivo" otherwise) and `description`.

Examples

```
estban_columns()[1:8, ]
v <- estban_verbetes()
v[v$code %in% c(160, 420, 432), c("name", "codes", "side")]
# Combined columns
v[v$n_codes > 1, c("codes", "n_codes")]
```

estban_download	<i>Download one ESTBAN month</i>
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Description

Downloads the monthly file for ref, trying each name variant from [estban_url\(\)](#), unzips it when needed and returns the path to the CSV in the cache directory. Months already in the cache are not downloaded again unless force = TRUE.

Usage

```
estban_download(
  ref,
  level = c("agencia", "municipio"),
  cache_dir = NULL,
  force = FALSE,
  timeout = 300,
  verbose = TRUE
)
```

Arguments

ref	Reference month as AAAAMM (integer, character or Date).
level	"agencia" (one row per bank branch, the default) or "municipio" (one row per institution and municipality).
cache_dir	Directory for downloaded files. See estban_cache_dir() for the resolution order; the default is a session folder under base::tempdir() .
force	Logical. Re-download even when the CSV is already cached.
timeout	Seconds allowed for one download (default 300).
verbose	Logical. Emit progress messages (default TRUE).

Value

The path to the cached CSV (invisibly NULL when the month is not published under any of the known names, with a warning).

Examples

```
# Needs network access to www.bcb.gov.br (about 2 MB); skipped when offline.
csv <- tryCatch(estban_download(202401, verbose = FALSE),
  error = function(e) NULL)
if (!is.null(csv)) {
  pe <- estban_read(csv, uf = "PE")
  nrow(pe)
}
```

estban_fetch	<i>Download and read a range of months</i>
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Description

Convenience wrapper around [estban_download\(\)](#) and [estban_read\(\)](#) for a range of reference months. Months that are not published are skipped with a warning, so the result may cover fewer months than requested; check `unique(result$ref)`.

Usage

```
estban_fetch(  
  start,  
  end = start,  
  uf = NULL,  
  level = c("agencia", "municipio"),  
  cache_dir = NULL,  
  clean_names = TRUE,  
  force = FALSE,  
  verbose = TRUE  
)
```

Arguments

start, end	Reference months as AAAAMM; end defaults to start.
uf	Optional character vector of two-letter state codes to keep (e.g. "PE" or <code>c("PE", "PB")</code>). NULL keeps every state.
level	"agencia" (one row per bank branch, the default) or "municipio" (one row per institution and municipality).
cache_dir	Directory for downloaded files. See estban_cache_dir() for the resolution order; the default is a session folder under <code>base::tempdir()</code> .
clean_names	Logical. Convert column names to snake_case with estban_clean_names() (default TRUE). With FALSE the original upper-case names are kept, including the leading # of #DATA_BASE.
force	Logical. Re-download even when the CSV is already cached.
verbose	Logical. Emit progress messages (default TRUE).

Value

A tibble with all requested months stacked (see [estban_read\(\)](#)).

Examples

```
# Needs network access; about 2 MB per month.
x <- tryCatch(estban_fetch(202401, 202403, uf = "PE", verbose = FALSE),
              error = function(e) NULL)
if (!is.null(x)) table(x$ref)
```

estban_flag_nonreport *Flag institution-months that were not reported*

Description

An institution can be present in an ESTBAN file with every account equal to zero in every branch and municipality of the extract. That is not a zero balance sheet: it is a **non-report** (the bank's return did not make it into that month's file). A documented case is Banco Santander in January to March 2025, zeroed in the whole country. Summing such rows into a municipal total makes credit and deposits collapse for three months and then jump back, which any time-series model reads as a real shock.

Usage

```
estban_flag_nonreport(df)
```

Arguments

df A tibble from `estban_read()` or `estban_fetch()` with clean names.

Details

`estban_flag_nonreport()` adds a logical column `nonreport` that is TRUE for every row of an institution-month whose accounts sum to zero across the whole table. Run it on the largest extract you have (ideally a state or the country), because the test is "zero everywhere in the data", and a single municipality can legitimately have a dormant branch.

Value

df with an extra logical column `nonreport`.

Examples

```
f <- system.file("extdata", "202401_ESTBAN_AG_sample.CSV", package = "estbanr")
x <- estban_read(f)
x <- estban_flag_nonreport(x)
table(x$nonreport)

# A synthetic non-report: zero every account of one bank in one month
y <- x
bank <- y$cnpj == y$cnpj[[1]]
y[bank, grep("^verbete_", names(y))] <- 0
table(estban_flag_nonreport(y)$nonreport, y$cnpj == y$cnpj[[1]])
```

estban_impute_nonreport

Impute non-reported institution-months

Description

Treats every institution-month flagged by `estban_flag_nonreport()` as missing and fills **interior** gaps by linear interpolation along the monthly series of each (institution, municipality, account). Gaps at the start or end of a series are left as NA (no extrapolation), so a bank that stopped reporting last month stays missing until the file is revised, instead of being invented.

Usage

```
estban_impute_nonreport(df)
```

Arguments

`df` A tibble from `estban_read()` or `estban_fetch()` covering several months (interpolation needs neighbours). The nonreport column is computed when absent.

Details

Rows are first summed to one row per institution and municipality (branches of the same bank in the same city are added), because that is the level at which the interpolation is meaningful and stable.

Value

A tibble with one row per `cnpj`, `codmun_ibge` and `ref`, the identification columns, the account columns (imputed where possible) and an integer column imputed with the number of accounts filled in that row. A message reports how many institution-months were treated.

Examples

```
# Three months of a two-bank, one-city extract, with bank B zeroed in the
# middle month. See vignette("nonreport-imputation") for the full story.
f <- system.file("extdata", "202401_ESTBAN_AG_sample.CSV", package = "estbanr")
m1 <- estban_read(f, uf = "PE")
m2 <- m1; m2$ref <- 202402L
m3 <- m1; m3$ref <- 202403L
verb <- grep("^verbete_", names(m1))
m2[m2$cnpj == "60746948", verb] <- 0 # Bradesco "vanishes" in February
m3[, verb] <- m3[, verb] * 1.10 # and everything grows 10% by March
x <- rbind(m1, m2, m3)
imp <- estban_impute_nonreport(x)
imp[imp$cnpj == "60746948" & imp$municipio == "CARUARU",
  c("ref", "verbete_160_operacoes_de_credito", "imputed")]
```

estban_read

Read an ESTBAN CSV file

Description

Reads a monthly ESTBAN file as published by the Central Bank: two title lines to skip, ; as separator, Latin-1 encoding, one row per bank branch (level = "agencia") or per institution and municipality (level = "municipio"). Account columns (VERBETE_*) are returned as doubles in Brazilian reais; codes are kept as character to preserve leading zeros.

Usage

```
estban_read(path, uf = NULL, clean_names = TRUE, ref = NULL)
```

Arguments

path	Path to a CSV downloaded with estban_download() (or any file with the same layout).
uf	Optional character vector of two-letter state codes to keep (e.g. "PE" or c("PE", "PB")). NULL keeps every state.
clean_names	Logical. Convert column names to snake_case with estban_clean_names() (default TRUE). With FALSE the original upper-case names are kept, including the leading # of #DATA_BASE.
ref	Optional reference month to record in the ref column. When NULL it is taken from the DATA_BASE column of the file.

Value

A tibble with a leading integer column ref (AAAAMM), the identification columns and one numeric column per account.

Examples

```
# A small real extract shipped with the package (PE and PB, January 2024)
f <- system.file("extdata", "202401_ESTBAN_AG_sample.CSV", package = "estbanr")
x <- estban_read(f)
x[, 1:8]

# Keep one state and the original column names
pe <- estban_read(f, uf = "PE", clean_names = FALSE)
names(pe)[1:10]
```

estban_url	<i>Candidate download URLs for one ESTBAN month</i>
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Description

The Central Bank has changed the file naming over the years, so for a given month more than one name may exist. The candidates are tried in order by [estban_download\(\)](#); the first that answers 200 wins.

Usage

```
estban_url(ref, level = c("agencia", "municipio"))
```

Arguments

ref	Reference month as AAAAMM (integer, character or Date).
level	"agencia" (one row per bank branch, the default) or "municipio" (one row per institution and municipality).

Value

A character vector of URLs, most likely first.

Examples

```
estban_url(202401)
estban_url("202401", level = "municipio")
```

Index

`base::tempdir()`, [3](#), [6](#), [7](#)

`estban_by_municipality`, [2](#)

`estban_cache_dir`, [3](#)

`estban_cache_dir()`, [6](#), [7](#)

`estban_clean_names`, [4](#)

`estban_clean_names()`, [7](#), [10](#)

`estban_columns`, [5](#)

`estban_download`, [6](#)

`estban_download()`, [7](#), [10](#), [11](#)

`estban_fetch`, [7](#)

`estban_fetch()`, [2](#), [8](#), [9](#)

`estban_flag_nonreport`, [8](#)

`estban_flag_nonreport()`, [9](#)

`estban_impute_nonreport`, [9](#)

`estban_impute_nonreport()`, [2](#)

`estban_read`, [10](#)

`estban_read()`, [2](#), [7–9](#)

`estban_url`, [11](#)

`estban_url()`, [6](#)

`estban_verbetes (estban_columns)`, [5](#)

`estban_verbetes()`, [4](#)